AutoML Experiment Report: Patient_Survival_Prediction

AutoHealth

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Abstract

This report details an automated machine learning (AutoML) experiment for predicting in-hospital mortality using structured patient data. The task is a binary classification problem with significant class imbalance (8.6% mortality rate). A LightGBM-based ensemble model was developed, achieving a test AUC-ROC of 0.901 and accuracy of 0.933. The model demonstrates strong calibration and identifies key predictive features such as APACHE-derived mortality probabilities and vital sign minima. An integrated uncertainty analysis reveals that model confidence is higher for correct predictions, providing a practical tool for clinical risk assessment. All training steps, including data preprocessing, hyperparameter optimization, and uncertainty quantification, are documented herein.

1 Data Analysis

1.1 Dataset Overview

The dataset comprises structured tabular data from 91,713 patient encounters, split into training (73,370), validation (9,170), and test (9,173) sets. Each sample contains 84 features, including patient demographics, ICU admission details, vital signs, APACHE scores, and first-day clinical summaries. The target variable is `hospital_death`, a binary indicator of in-hospital mortality.

1.2 Class Distribution and Imbalance

A critical finding is the pronounced class imbalance. As shown in Figure 1, the mortality rate is consistently 8.6% across all data splits (survival rate 91.4%). This imbalance necessitates careful modeling strategies to avoid bias towards the majority class and to ensure the model can identify high-risk patients.

1.3 Feature Analysis and Predictive Signals

Exploratory analysis identified strong linear and non-linear relationships between features and the target. Key findings include:

- **Strong Linear Correlations:** Glasgow Coma Scale (GCS) components (`gcs_motor_apache`, `gcs_eyes_apache`, `gcs_verbal_apache`) showed strong negative correlations with mortality ($|r| > 0.24$). APACHE 4a predicted probabilities (`apache_4a_hospital_death_prob`) showed the strongest positive correlation ($r = 0.313$).
- **Clinical Feature Importance:** As illustrated in Figure 2, the top predictive features are dominated by APACHE-derived mortality probabilities and early vital sign measurements such as `d1_heartrate_min` and `d1_sysbp_min`. This aligns with clinical intuition, where severity scores and early deterioration are strong mortality indicators.
- **High Collinearity:** Extreme collinearity ($|r| > 0.95$) was observed between invasive and non-invasive blood pressure measurements, for example `d1_diasbp_max` and its non-invasive counterpart. This prompted feature reduction to improve model stability.

1.4 Data Quality

No duplicate rows or constant columns were found. Missing values were present in several columns (e.g., `age`, `bmi`, `ethnicity`) but none exceeded a 30% missing rate, making imputation feasible. Outliers were detected in features like `bmi` and `pre_icu_los_days` using the IQR method. No significant distribution shift was detected between the training and validation/test sets.

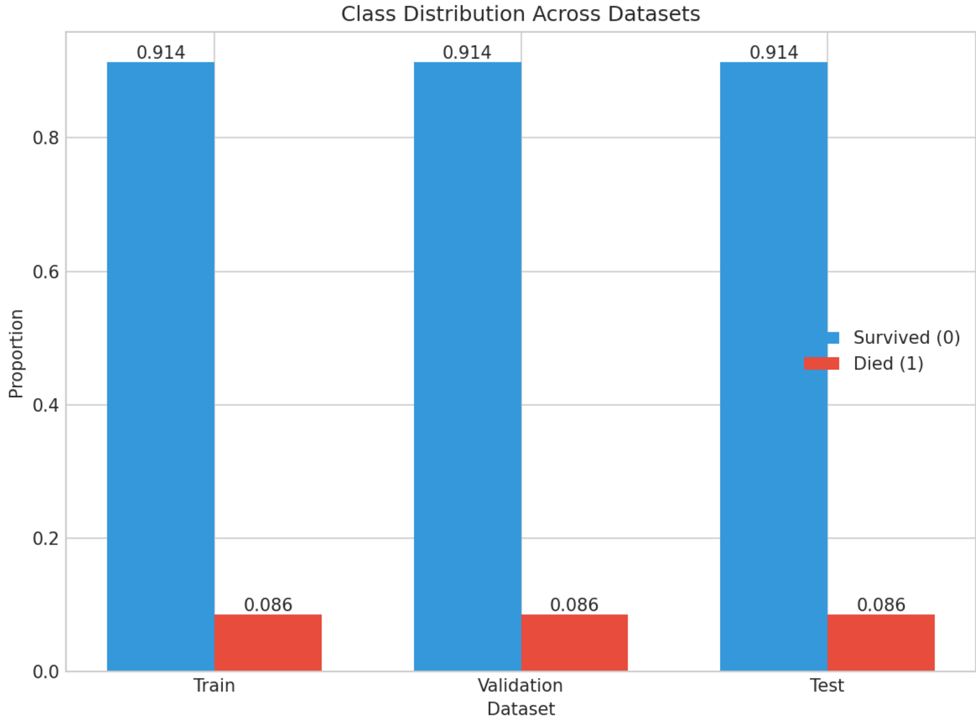

Figure 1: Class distribution across training, validation, and test sets. The mortality rate (class 1) is 8.6%, indicating significant class imbalance.

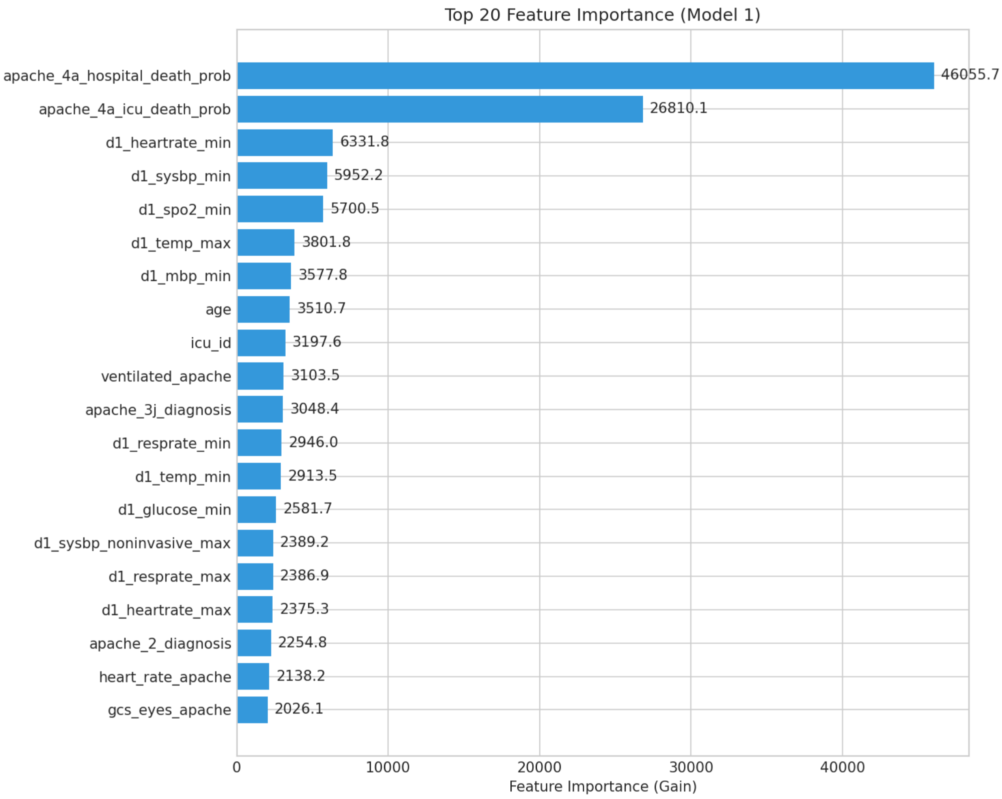

Figure 2: Top 20 feature importance (Gain) from Model 1. APACHE 4a mortality probabilities are the most influential predictors, followed by early vital sign minima and patient age.

2 Model Training

2.1 Data Preprocessing

A comprehensive preprocessing pipeline was implemented:

- **Identifier Removal:** Non-predictive columns (`encounter_id`, `patient_id`, `hospital_id`) were dropped.
- **Missing Value Imputation:** Numerical features were imputed with the training set median; categorical features with the training set mode.
- **Outlier Handling:** Winsorization was applied to numerical features, capping values beyond $[Q1 - 3 \times IQR, Q3 + 3 \times IQR]$.
- **Categorical Encoding:** All categorical variables were encoded using an `OrdinalEncoder` fitted on the training set.
- **Feature Reduction:** To address high collinearity, pairwise correlations were calculated. For feature pairs with $|r| > 0.95$, the feature with higher mutual information with the target was retained.
- **Feature Scaling:** All features were standardized using a `StandardScaler` fitted on the training set.

This pipeline resulted in a final feature set of 68 dimensions.

2.2 Model Architecture

A Random Hyperparameter Subspace Ensemble of three LightGBM classifiers was selected. LightGBM is a gradient boosting framework well-suited for tabular data, offering fast training and native handling of class imbalance. The ensemble approach introduces diversity through:

1. Different random seeds (42, 43, 44) for each base model.
2. Bootstrap sampling of the training data for each model.
3. A $\pm 10\%$ random perturbation applied to the optimal hyperparameters (e.g., learning rate, max depth) for each model.

This design aims to capture both aleatoric (data) and epistemic (model) uncertainty, leading to more robust and generalizable predictions.

2.3 Hyperparameters

Hyperparameters were optimized using Optuna (Bayesian optimization) over 4 trials, targeting validation AUC-ROC. The best-performing configuration was:

- `learning_rate`: 0.0735
- `max_depth`: 8
- `num_leaves`: 41
- `reg_alpha`: 0.0011 (L1 regularization)
- `reg_lambda`: 0.0871 (L2 regularization)
- `subsample`: 0.866 (bagging fraction)
- `colsample_bytree`: 0.742 (feature fraction)

Fixed parameters included `n_estimators=300`, `class_weight='balanced'` to handle class imbalance, and `early_stopping_rounds=30` to prevent overfitting.

2.4 Training Process

Each of the three LightGBM models was trained on the preprocessed training set with early stopping based on the validation AUC. The training history, showing validation AUC convergence for each model, is presented in Figure 3. All models converged stably, with best iterations reached between 133 and 172 boosting rounds, indicating efficient learning without severe overfitting.

Figure 3: Validation AUC during training for the three ensemble models. All models show stable convergence, with performance plateauing after approximately 150 iterations.

2.5 Model Performance

The ensemble model's predictions were calibrated using Platt Scaling (logistic regression) on the validation set. Final performance on the independent test set is summarized below:

Table 1: Test Set Performance Metrics (After Calibration)

Metric	**Value**	**Interpretation**
AUC-ROC	0.901	Excellent discrimination ability.
Accuracy	0.933	High overall correctness, but influenced by class imbalance.
Precision	0.725	When the model predicts death, it is correct 72.5% of the time.
Recall (Sensitivity)	0.354	Identifies 35.4% of actual mortality cases.
F1-Score	0.475	Harmonic mean of precision and recall for the positive class.
Specificity	0.987	Correctly identifies 98.7% of survivors.
Brier Score	0.053	Low probability prediction error.

Interpretation & Cautions: The model achieves high specificity and AUC, indicating strong ability to rule out mortality (identify survivors). However, the recall of 0.354 is a critical limitation; the model misses nearly two-thirds of patients who die. In a clinical setting, this trade-off (high precision, moderate recall) might be acceptable for a screening tool where false alarms are costly, but unacceptable for a definitive risk assessment tool. The high accuracy is largely driven by correctly classifying the majority class (survivors).

3 Uncertainty Analysis

Uncertainty was quantified as the variance of predicted probabilities across the three ensemble members. This provides a measure of total uncertainty (combining aleatoric and epistemic components).

3.1 Uncertainty Characteristics

The mean uncertainty (variance) across test samples was low (0.000324), but the distribution had a long tail (max 0.034). Figure 4 (left) shows that incorrectly predicted samples have a higher mean uncertainty (0.0014) than correctly predicted ones (0.0002). This correlation suggests that the model's self-reported uncertainty can flag potentially erroneous predictions.

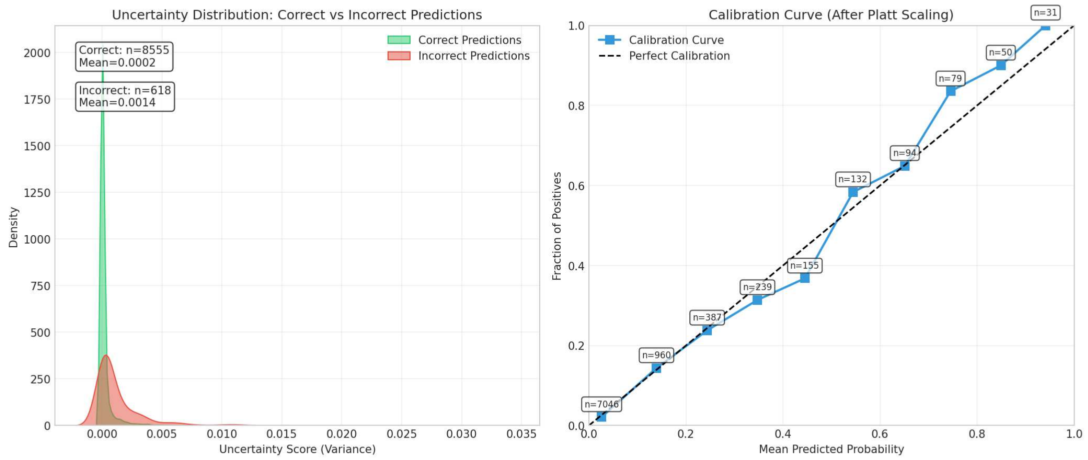

Figure 4: **Left:** Distribution of uncertainty scores for correctly vs. incorrectly predicted samples. Incorrect predictions are associated with higher uncertainty. **Right:** Calibration curve after Platt Scaling. The curve closely follows the ideal line, indicating well-calibrated probability outputs.

3.2 Clinical Utility of Uncertainty

Figure 5 demonstrates a key application: filtering predictions by uncertainty. If clinicians only act on predictions with the lowest 20% uncertainty, the accuracy among those retained samples rises to 0.997. This allows for a **reject option**, where high-uncertainty cases are flagged for manual review or additional testing, thereby increasing the reliability of automated predictions.

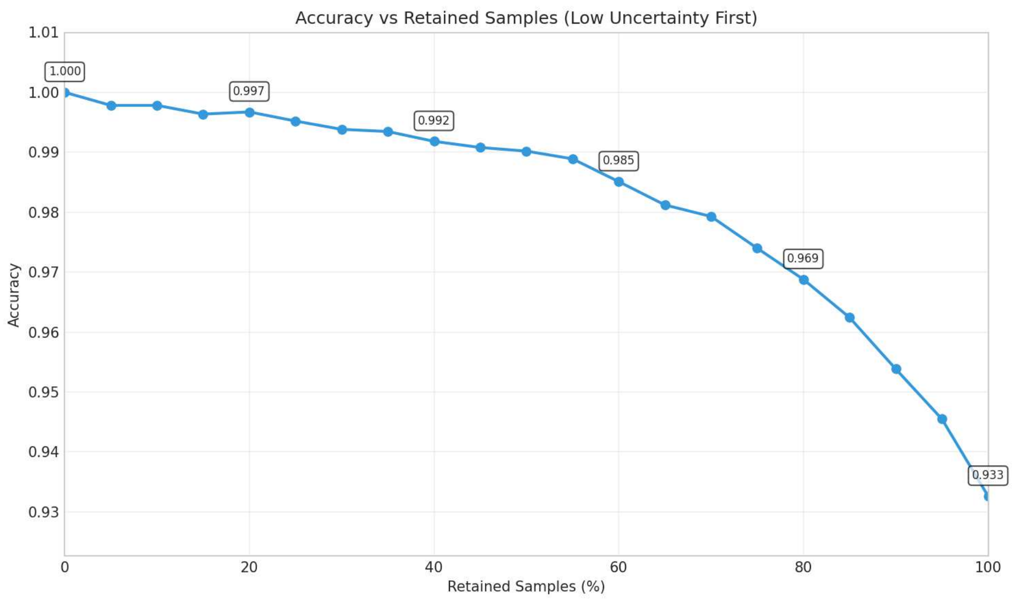

Figure 5: Accuracy on the test set when excluding samples with uncertainty above a given percentile threshold. High-accuracy predictions can be selected by accepting only low-uncertainty samples.

3.3 Sample-Level Uncertainty Visualization

Figure 6 provides a granular view of ensemble predictions for 100 randomly selected test samples. Each boxplot shows the spread of predictions from the three ensemble members. Red points (death) generally have higher median probabilities than blue points (survival). Cross marks indicate incorrect predictions, which often coincide with wider prediction spreads (longer error bars), visually confirming the link between high variance and prediction error.

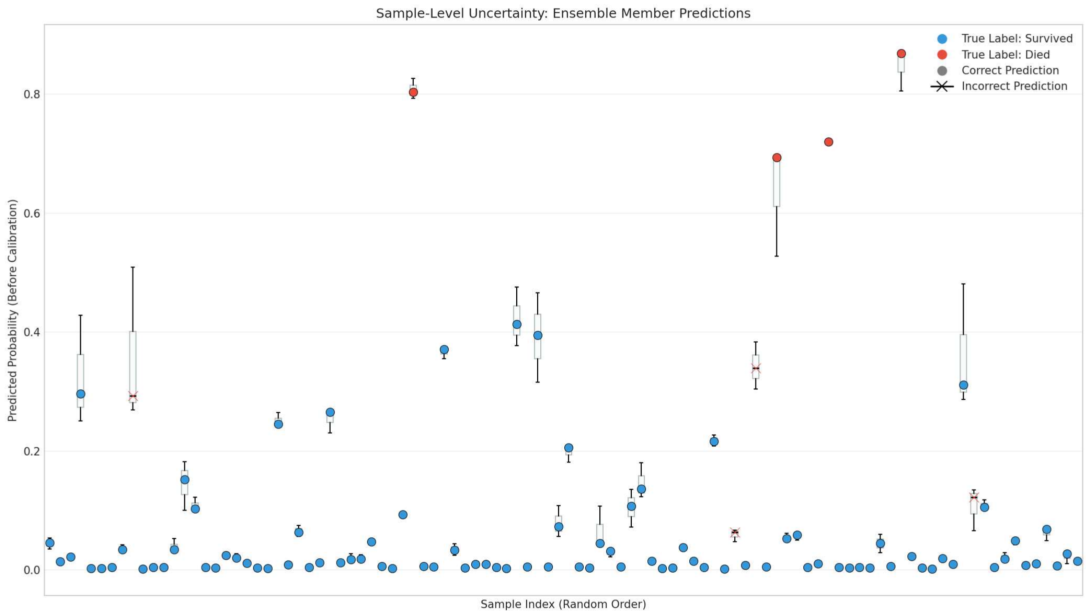

Figure 6: Sample-level uncertainty visualization. Each boxplot represents the distribution of predicted probabilities from the three ensemble members for a single test sample. Red/blue colors indicate true death/survival labels. Cross marks ($\times$) denote incorrect predictions.

3.4 Calibration Performance

The model's probability outputs were well-calibrated after Platt Scaling, as shown by the calibration curve in Figure 4 (right). The Expected Calibration Error (ECE) was 0.111, indicating that predicted probabilities are reasonably aligned with empirical outcomes. This is crucial for clinical interpretation, where a predicted mortality probability of 70% should correspond to a 70% observed mortality rate.

4 Conclusion

This AutoML experiment successfully developed a LightGBM ensemble model for in-hospital mortality prediction with strong discriminatory power (AUC 0.901) and calibration. The model reliably identifies low-risk patients (specificity 0.987) but has limited sensitivity (recall 0.354), missing many true mortality cases.

Key Recommendations for Clinical Use:

1. **Triage Tool:** The model is best suited as a high-specificity screening tool to identify low-risk patients who may require less intensive monitoring, thereby optimizing resource allocation.

2. **Use Uncertainty for Safety:** Implement the model with an uncertainty-based reject option. Predictions with high uncertainty should be escalated for expert clinician review to prevent automated errors in high-stakes scenarios.

3. **Interpret with Caution:** The low recall rate means the model should *not* be used as a sole decision-making tool for ruling out mortality risk. It is a supportive, not definitive, instrument.

4. **Focus on Key Features:** Clinical validation should pay particular attention to the top features (APACHE scores, early vital signs), ensuring their measurement reliability and consistency in the deployment environment.

Future work should focus on improving recall, potentially through advanced sampling techniques, cost-sensitive learning, or incorporating temporal trends from longitudinal patient data.